Suppose male and female earnings are given by the following equations

Ym = 5000 + 100Pm

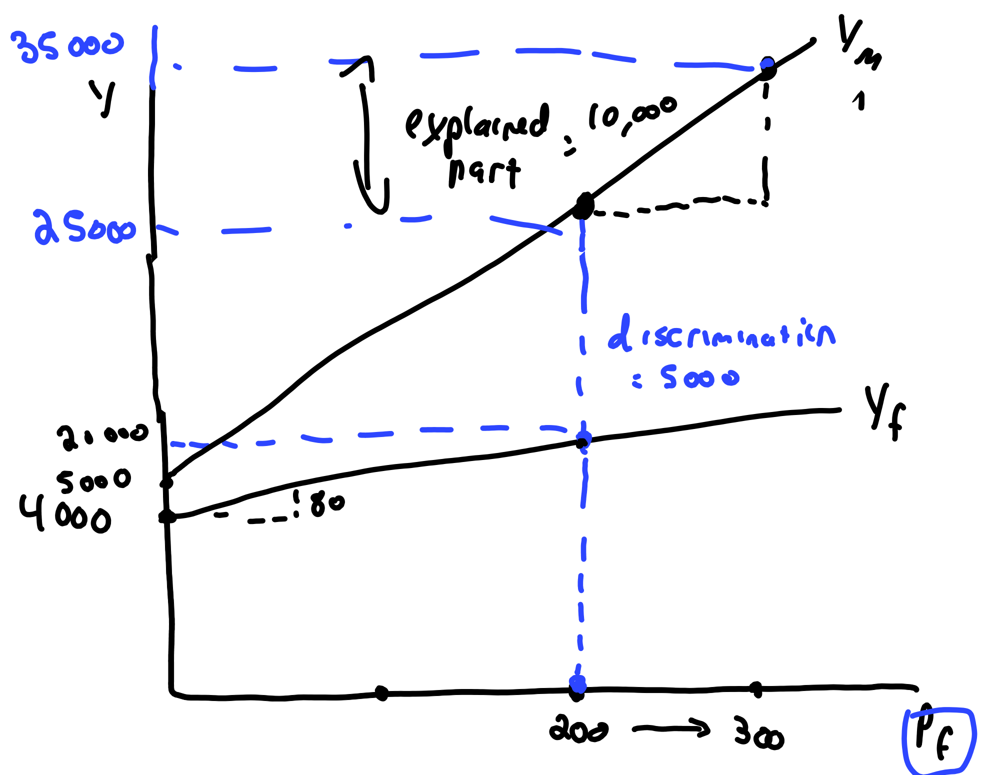
Yf = 4000 + 80Pf

Where P is job evaluation "points." Assume the average job evaluation score for male jobs is 300 and 200 for female jobs.

a) What is the average pay in male-dominated jobs and in female dominated jobs?

$$Y_m = 5000 + 100(300)$$
$$= 5000 + 30000$$
$$= 35000$$

$$Y_f = 4000 + 80(200)$$
$$= 4000 + 16000$$
$$= 20000$$



b) Use the Oaxaca decomposition to determine the average pay differential between male and female dominated jobs into two components: a portion attributable to the differences in job evaluation points, and a portion attributable to differences in pay for the same job evaluation points.

$$Y_m = \beta_{om} + \beta_{1m} P_m$$

$$Y_f = \beta_{of} + \beta_{1f} P_f$$

$$Y_m - Y_f = \beta_{om} - \beta_{of} + \beta_{1m} P_m - \beta_{1f} P_f + \beta_{1m} P_f - \beta_{1m} P_f$$

$$= \underbrace{\beta_{om} - \beta_{of} + P_f(\beta_{1m} - \beta_{1f})}_{\text{unexplained component}} + \underbrace{\beta_{1m}(P_m - P_f)}_{\text{explained component}}$$

$$= 5000 - 4000 + 200(100 - 80) + 100(300 - 200)$$

$$= 5000 + 10\,000$$

So, $\frac{1}{3}$ of the gap is unexplained, $\frac{2}{3}$ explained

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alternative: what would women earn with male returns?
(i.e. male equation)

$$Y_f^* = 5000 + 100(200)$$
$$= \beta_{om} + \beta_{1m} P_f \qquad \text{compare to } Y_f = 20{,}000$$
$$= 25000$$

remainder is \$10,000, which is due to observed differences

c) Use the same decomposition, but this time evaluate the differences in the job evaluation points according to the female pay for such points

$$Y_m - Y_f = \beta_{0m} - \beta_{0f} + \underline{\beta_{1m} P_m} - \beta_{1f} P_f + \beta_{1f} P_m - \underline{\beta_{1f} P_m}$$

$$= \underbrace{\beta_{0m} - \beta_{0f} + P_m(\beta_{1m} - \beta_{1f})}_{\text{unexplained}} + \underbrace{\beta_{1f}(P_m - P_f)}_{\text{explained}}$$

$$= 1000 + 300(20) + 80(100)$$

$$= \boxed{\begin{matrix} 7000 \\ \text{unexplained} \end{matrix}} + \boxed{\begin{matrix} 8000 \\ \text{explained} \end{matrix}}$$

alternative: what if men followed the female equation

$$Y_m^* = \underset{\beta_{0f}}{4000} + \underset{\beta_{1f} P_m}{80(300)}$$

$$= 4000 + 24000$$

$$= 28\,000$$

compare to $Y_m = 35000$

difference of 7000 which is unexplained

compare to $Y_f = 20\,000$

difference is 8 000 explained

1) The sources of discrimination include all of the following, with the exception of:

A) employers.

B) Customers who are served by the workers.

C) male co-workers.

D) systemic human resource management practices in the labour market.

E) productivity differences.

2) The demand theory of discrimination focuses on which of the following?

A) The demand for female labour is reduced relative to the demand for equally productive male labour.

B) The demand for female labour relies on the preferences of the recruitment and hiring committee of each firm.

C) Females tend to be less aggressive than males when they are negotiating wages with their employer.

D) Females are restricted from entering certain occupations, and are thus crowded into other occupations.

E) Women have a higher labour force participation rate than men.

3) Within the framework of the Oaxaca decomposition, the basic approach to analyzing whether there exists wage discrimination against women is to:

A) compare the actual mean wage of women to the actual mean wage of men.

B) compare the actual wage of women to the predicted wage that they would earn given female attributes and coefficients from the male equation.

C) compare the actual wage of women to the predicted wage that they would earn given male attributes and coefficients from the male equation.

D) compare the actual wage of women to the predicted wage that they would earn given female attributes and coefficients from the female equation.

E) search for anecdotal cases of low-paid women that appear to be affected by discrimination.

4) The supply theory of discrimination focuses on which of the following?

   A) The supply of female labour is reduced relative to the demand of female labour.

   B) The supply of female labour relies on the preferences of the hiring committee of each firm.

   C) The supply of female labour to certain occupations or industries is increased as a result of discrimination.

   D) Women have a higher labour force participation rate than men

   E) Females receive lower salaries for the typical "male-type" jobs.

Answer Key

1) E
2) A
3) B
4) C